

Data Cleaning & Reporting Automation

Executive Pipeline Report | Generated on August 26, 2026 at 22:56:45

1. Executive Summary & Health Scorecard

Health Metric	Raw Ingestion	Cleaned Result	Pipeline Impact
Data Quality Score	74.8/100 [B (Acceptable)]	100.0/100 [A+ (Excellent)]	+25.2 Quality Gain
Total Rows	27	24	3 Duplicates Removed
Total Columns	7	9	Standardized snake_case
Missing Values	9	0	100% Imputed / Resolved

2. Key Performance Indicators (KPIs)

KPI Indicator	Computed Value	Description
Total Records	24	Total active rows in processed dataset
Total Revenue / Sales	\$18,442,434.00	Cumulative sum of 'gross_revenue'
Average Order Value	\$768,434.75	Average ticket size from 'gross_revenue'
Max Transaction	\$1,450,130.00	Highest recorded value in 'gross_revenue'
Total Gross Profit	\$9,162,666.00	Gross profit calculated from 'gross_profit'
Average Margin	48.1%	Mean profit margin percentage

3. Visual Performance Analysis

Visual chart rendering skipped: name 'np' is not defined

4. Automated Insights & Findings

- **Insight:** Overall Data Health Score improved from 74.8/100 [B (Acceptable)] to 100.0/100 [A+ (Excellent)].
- **Insight:** Resolved 9 missing values across 2 columns using statistical imputation strategies.
- **Insight:** Eliminated 3 exact duplicate rows, preventing skewed metrics and double-counting.
- **Insight:** Successfully standardized data types for 4 columns (including currency symbol stripping, date parsing, and boolean normalization).
- **Insight:** Standardized 7 column headers into consistent snake_case format for SQL/pipeline compatibility.
- **Insight:** Total cumulative revenue / sales volume stands at \$18,442,434.00 across 24 transactions.
- **Insight:** Average Order Value (AOV) was calculated at \$768,434.75 per transaction.
- **Insight:** Top contributing category is 'General & Admin' accounting for 30.4% of total volume (\$5,610,515.00).
- **Insight:** Latest period performance demonstrated a growth rate of +52.1%.

5. Strategic Recommendations

- **Action:** Implement frontend form validation at data entry to mandate required fields and eliminate downstream missing values.
- **Action:** Enforce database UNIQUE constraints on transactional ID fields to prevent duplicate records at ingestion.

- **Action:** Enforce strict schema typing at the ingestion pipeline to avoid storing raw currency symbols and unformatted strings in numeric fields.
- **Action:** Evaluate marketing and promotional strategies for lower-performing category 'Operations' (20.1% share) to balance portfolio revenue.
- **Action:** Schedule automated daily/weekly runs of this cleaning pipeline to monitor data health drift over time.